

Biomaterials

生物材料

南方科技大学材料科学与工程系2024年春季课程

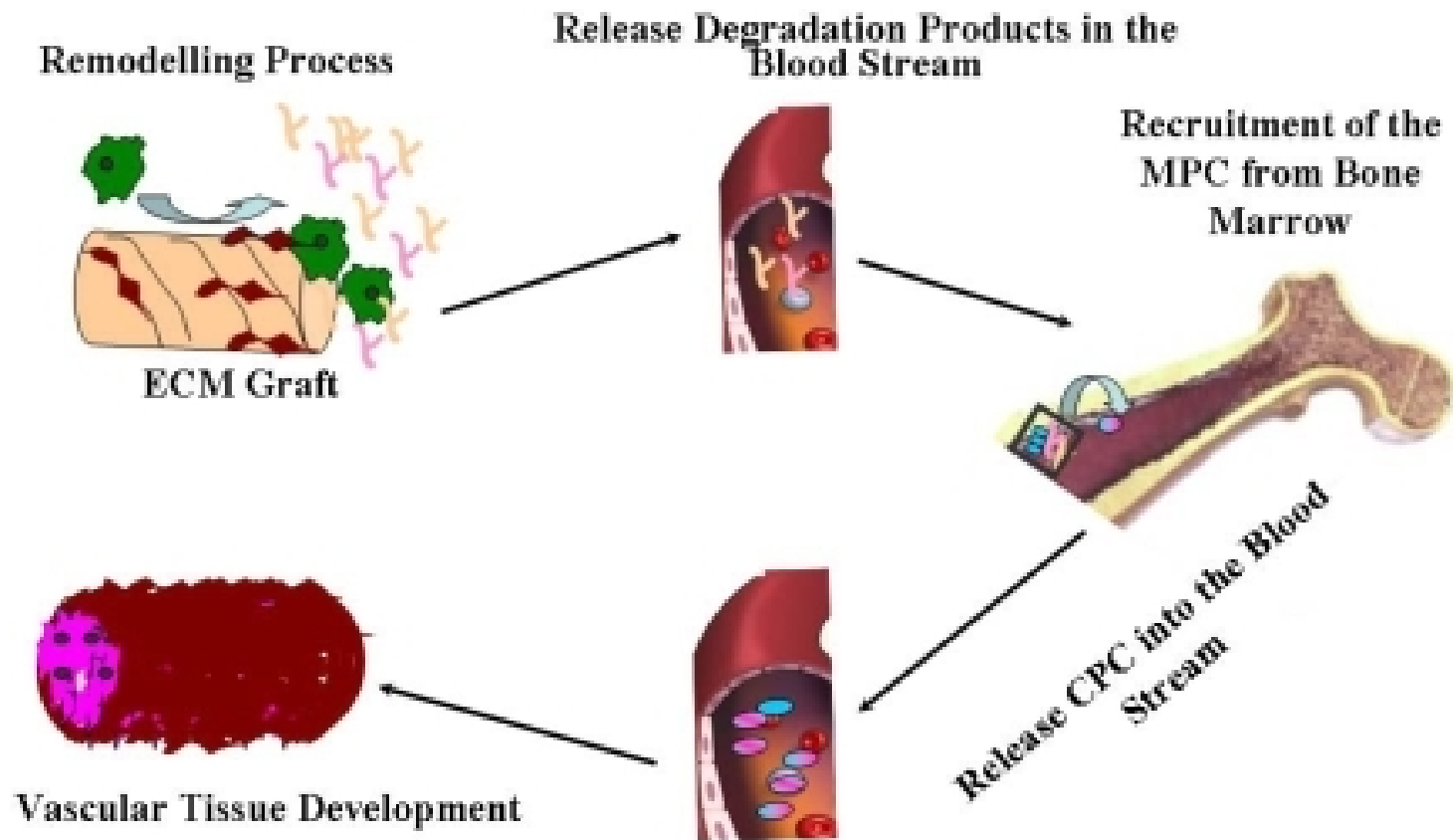
Host Reactions to Biomaterials and Their Evaluation

- A: Implantation and Inflammation, and the foreign-body response
- B: The Immune response to foreign materials
- C: Wound healing and the presence of biomaterials
- D: Blood coagulation and blood-materials interactions
- E: Biofilms, biomaterials, and device-related infections

Implantation & Inflammation

- Innate and Acquired Immunity 先天和后天免疫
- Physiological Response to Acute Inflammation
- Signs of Inflammation and their Cause
- Actions of Neutrophils
- Actions of Macrophages

Host Response to Material Implantation



-Macrophage



-Multipotent Progenitor cell (MPC)



-Circulating Progenitor cells (CPC)



-Degradation Products of ECM material



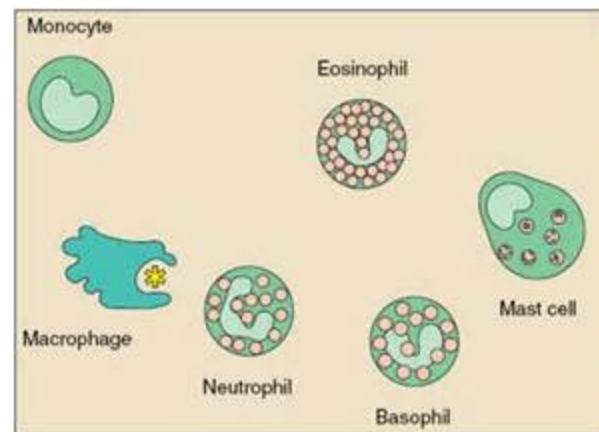
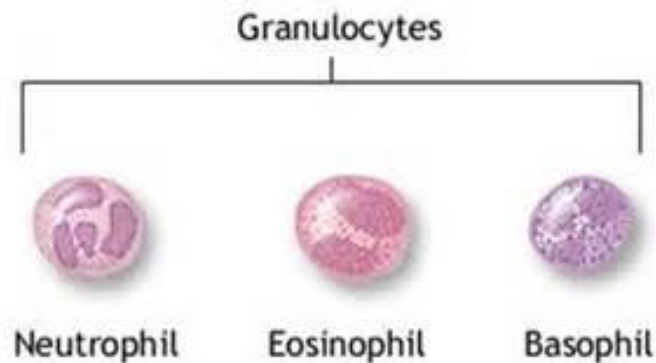
-Vascular Smooth Muscle Cells



-Vascular Endothelial cell

Types of Immunity

- Innate (nonspecific) – 1st line of defense
 - Anatomic barriers (e.g, skin and mucous membranes)
 - Physiologic barriers (body temp., low pH in stomach)
 - Phagocytic cells (granulocytes)
 - Inflammation
- Acquired (specific):
 - Activation of white blood cells (lymphocytes)
 - Develops following exposure to certain *pathogens*



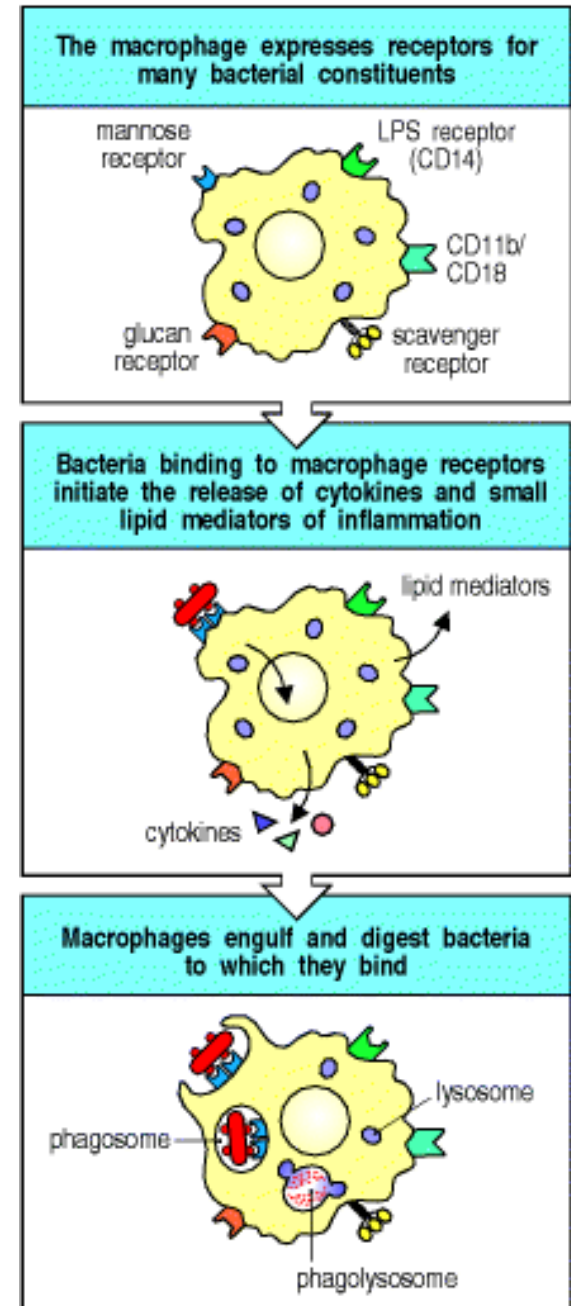
Granulocytes are a category of white blood cells characterized by the presence of granules in their cytoplasm. They are also called polymorphonuclear leukocytes (PMN, PML, or PMNL) because of the varying shapes of the nucleus, which is usually lobed into three segments. This distinguishes them from the mononuclear agranulocytes. In common parlance, the term **polymorphonuclear leukocyte** often refers specifically to neutrophil granulocytes, the most abundant of the granulocytes; the other types (eosinophils, basophils, and mast cells) have lower numbers. Granulocytes are produced via granulopoiesis in the bone marrow.

Innate Immune response

- If a pathogen breaches the epithelium, then the innate immune response begins.
- The cells of the immune system determine “self” from “non-self” by recognizing molecules on the microbe surface.
- Macrophages are immune cells (phagocytes) that reside within the tissue. Neutrophils are phagocytes that reside in the blood but can extravasate into tissue during inflammation.
- There are circulating proteins, called complement, that either kill microbes or mark them for effective phagocytization.

CD14（cluster of differentiation 14），即LPS

（Lipopolyssacharide）受体，最初是一种存在于单核细胞、巨噬细胞等细胞表面的白细胞分化抗原，由TODD于1981年首次从人单核细胞表面发现。



单核细胞（英语：Monocyte）是人体免疫系统的一种白细胞。当它们从骨髓进入血液时仍然是尚未成熟的细胞。其在人体免疫系统内有两种作用：一，补充正常状态下的巨噬细胞和树状细胞；二，在有炎症信号下，单核细胞会在8到12小时快速聚集到感染组织，并分化出巨噬细胞和树状细胞产生免疫反应。其细胞核常偏位，呈多形性，如卵圆形、肾形、马蹄形、不规则形等，常有折叠感；染色质呈疏松网状，着色较浅。与其他血细胞比较，单核细胞内含有更多的非特异性脂酶，并且具有更强的吞噬作用。胞质较多，嗜碱性，但因含大量细小的嗜天青颗粒而染成灰蓝色，颗粒含过氧化物酶。单核细胞在血液中停留2-3天后迁移到周围组织中，细胞体积继续增大，直径可达50-80 μm ，细胞内所含的溶酶体颗粒和线粒体的数目也增多，成为成熟的细胞。激活了的单核细胞和组织巨噬细胞能生成并释放多种细胞毒、干扰素和白细胞介素，参与机体防卫机制，还产生一些能促进内皮细胞和平滑肌细胞生长的因子。在炎症周围单核细胞能进行细胞分裂，并包围异物。

多核体apocyte

具有二个以上细胞核的原生质团，又称多核细胞或共浆体。它形成的原因可以区分为：（1）细胞融合，由于细胞质混合的结果产生的合胞体；（2）在单核细胞内细胞核进行了分裂，但细胞分裂与其结果不相伴所形成的合胞体是有区别的。在高等植物的乳管以及变形虫形的绒毡层细胞、动物的肌肉、结缔组织中，可以看到（1）的例子；某种高等植物在胚发育的某个时期，车轴藻类的节间细胞，粘菌类的变形体，管藻类的营养体等细胞中，可以看到（2）的例子。

Host Defense

- The body is under constant attack by microorganisms in the environment.
- Pathogen : an infectious agent that causes disease
- Infectious disease occurs when a microorganism succeeds in evading or overwhelming host defenses to establish a local site of infection and replication. In order for a pathogen to enter the body it must first overcome the epithelium and then the **innate immune response**.

First Line of Defense

- **Epithelial Tissue**
 - covers the whole surface of the body
 - made up of closely packed cells
 - can be divided into simple or stratified
 - interior epithelium covered with a mucus layer

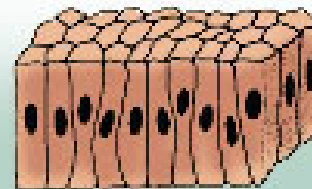
Types of Epithelium



Simple squamous

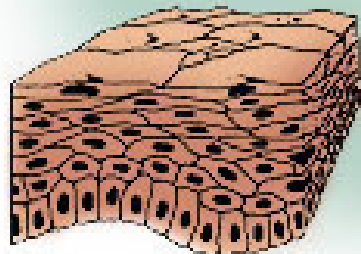
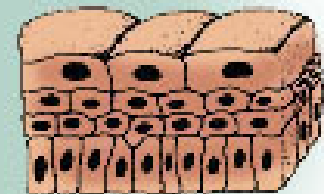


Simple cuboidal

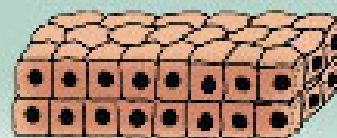


Simple columnar

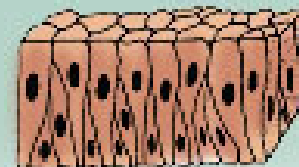
Transitional



Stratified squamous



Stratified cuboidal



Pseudostratified columnar

Types of Leukocytes 白细胞类型

- Granulocytes: Includes *neutrophils* (中性粒细胞), *eosinophils* (嗜酸性细胞) and *basophils* (嗜硷性细胞):
 - Granular appearance
 - Nuclei have multiple lobes
 - Phagocytose foreign invaders
 - Aid in inflammatory response

Phagocytosis: the cellular process of engulfing solid particles by the cell membrane to form an internal phagosome, which is a food vacuole. The phagosome is usually delivered to the lysosome, an organelle involved in the breakdown of cellular components, which fuses with the phagosome. The contents are subsequently degraded and either released extracellularly via exocytosis, or released intracellularly to undergo further processing.

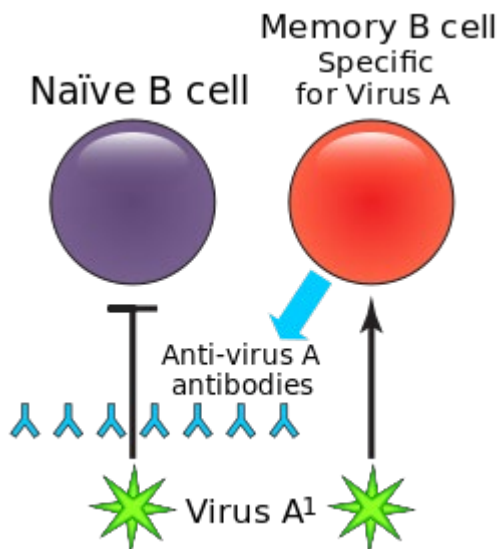
Types of Leukocytes, Continued

- Monocytes:
 - Large phagocytic capability
 - Play key roll in inflammation response
- Lymphocytes/plasma cells:
 - Includes T- and B-cells
 - Part of acquired immune response
 - Lymphocytes have *memory cells*, give rapid response when exposed to same pathogen
 - Lymphocytes also have *effector cells*, producing antibodies and try to remove foreign invaders
- Megakaryocytes (巨核细胞):
 - Found only in bone marrow
 - Produces platelets that participate in clotting

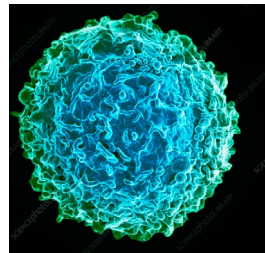
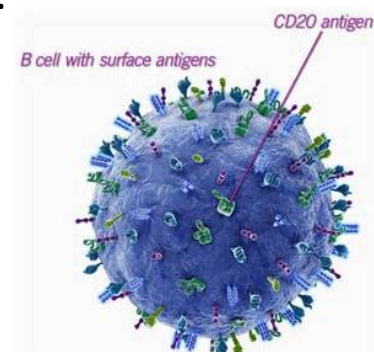
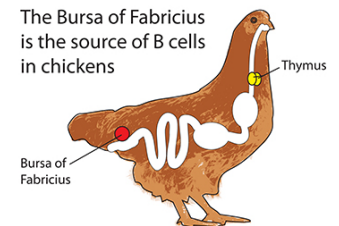
B cells or **B lymphocytes** are a type of [lymphocyte](#) in the [humoral immunity](#) of the [adaptive immune system](#). B cells can be distinguished from other lymphocytes, such as [T cells](#) and [natural killer cells](#) (NK cells), by the presence of a [protein](#) on the B cell's [outer surface](#) known as a [B cell receptor](#) (BCR). This specialized [receptor protein](#) allows a B cell to bind to a **specific** [antigen](#). In birds, B cells mature in the [bursa of Fabricius](#) (the name of “B” cells comes from the name of this organ).^[1] In mammals, immature B cells are formed in the [bone marrow](#).^[2]

[Bursa of Fabricius](#): 法氏囊;法布利氏囊;腔上囊

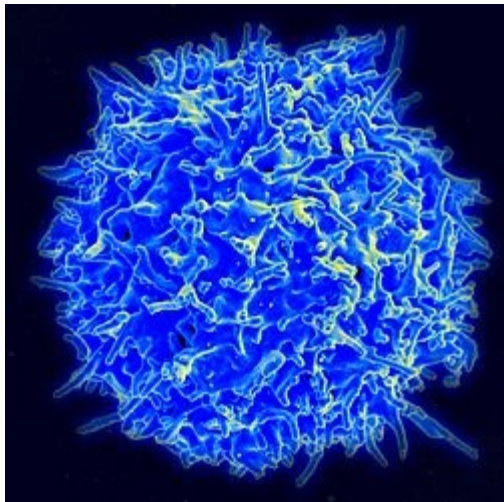
The principal functions of B cells are to make [antibodies](#) to help [antigens](#), to perform the role of [antigen-presenting cells](#) (APCs), and to develop into memory B cells after activation by antigen interaction. B cells also release [cytokines](#) (proteins), which are used for signaling immune regulatory functions.^[3]



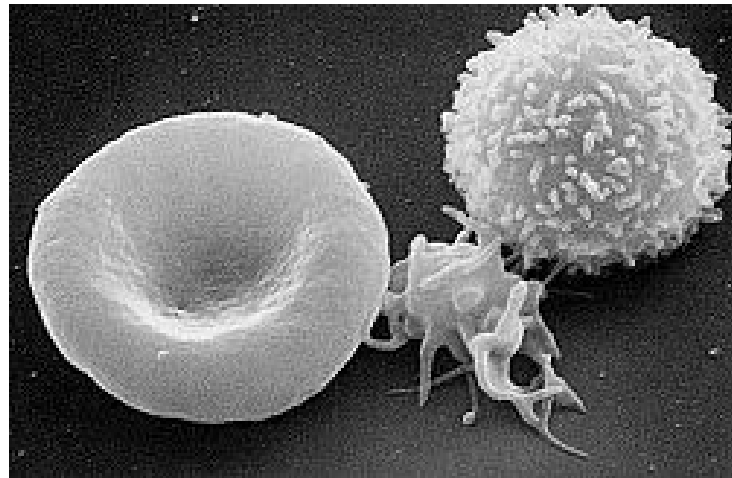
The cells of the immune system that make antibodies to invading pathogens such as viruses. They form memory cells that remember the same pathogen for faster antibody production in future infections.



T cells or **T lymphocytes** are a type of [lymphocyte](#) (in turn, a type of [white blood cell](#)) that plays a central role in [cell-mediated immunity](#). They can be distinguished from other lymphocytes, such as [B cells](#) and [natural killer cells](#) (NK cells), by the presence of a [T-cell receptor](#) (TCR) on the cell surface. They are called T cells because they mature in the [thymus](#) (although some also mature in the [tonsils](#)).^{[1][2]} The several subsets of T cells each have a distinct function.



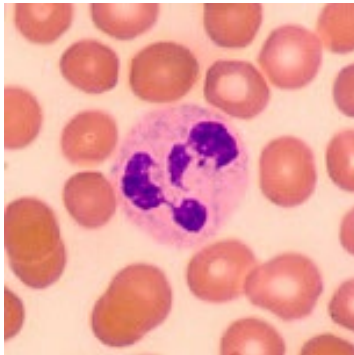
Scanning electron micrograph of a human T cell



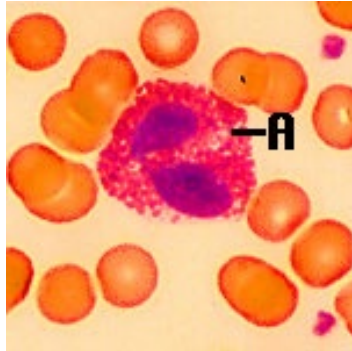
Scanning electron micrograph of T lymphocyte (right), a platelet (center) and a red blood cell

Intravascular Cells

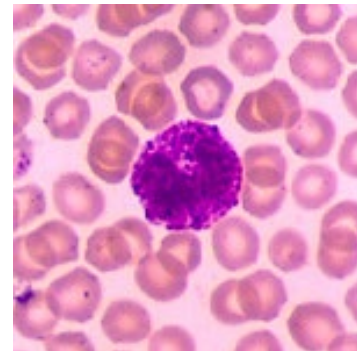
- white blood cells



neutrophil

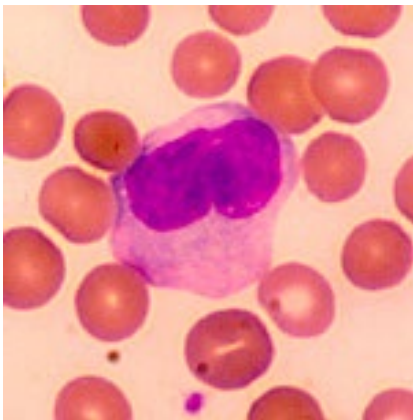


eosinophil

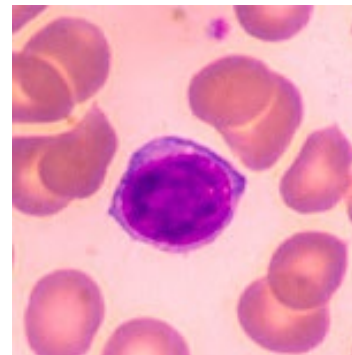


basophil

— Granulocytes



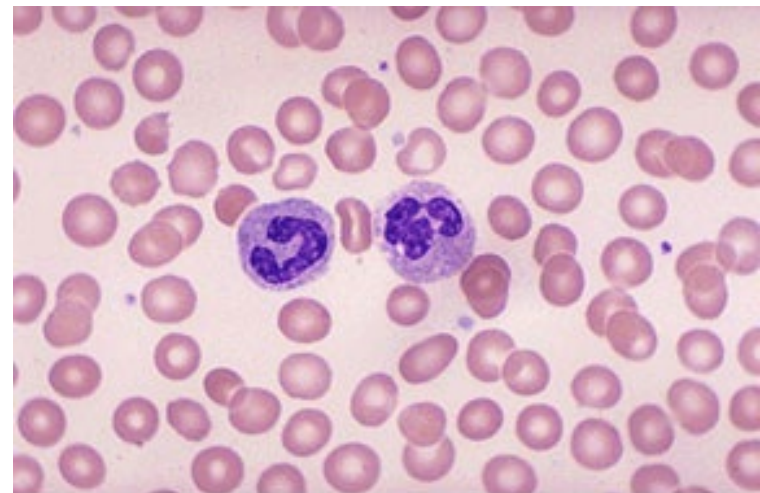
monocyte



lymphocyte

Wright and Giemsa stains are Romanowsky stains used to stain peripheral blood and bone marrow smears. The most important components of these stains are **oxidized methylene blue, azure B and eosin Y dyes**. The eosin Y dye stains the cytoplasm of cells an orange to pink color. The methylene blue and azure B dyes stain the nucleus varying shades of blue to purple. Romanowsky stains are used to perform differential white blood cell counts and to study red blood cell morphology. Abnormal granulocyte, lymphocyte or monocyte cell counts may be used to facilitate the diagnosis of diseases such as leukemia or bacterial infections.

White blood cells stained with Wright's stain:



Neutrophils Stained with Wright Stain

Leukocyte Formation

- Leukocytes formed in pluripotent hematopoietic (多能造血细胞) cells in bone marrow
- Granulocytes, monocytes, megakaryocytic form in bone marrow
- Lymphocytes form in bone marrow and mature in lymphoid tissues (lymph glands, spleen, thymus, tonsils (淋巴结, 脾脏, 胸腺, 扁桃体))

Life Span of Leukocytes

- Only present in blood as needed
- Granulocytes:
 - 4-8 hrs. in blood
 - 4-5 days in tissue
- Monocytes:
 - 10-20 hrs. in blood
 - Migrate to tissue, becoming “tissue macrophages”
- Tissue macrophages
 - Longer lasting and more powerful than monocytes
 - Can live months to years
 - Provide continual resident defense against infection
- Lymphocytes
 - Resident in lymphoid tissue
 - Enter blood stream for only a few hours
 - Re-enter the lymph system
 - Can continue in this loop for months to years

Inflammatory Response

- Pathogen recognition and tissue damage begin an inflammation response. This is characterized by :
 - swelling
 - pain
 - redness
 - heat
- Inflammation allows for neutrophil and plasma protein extravasation. Both of these effects aids the immune response.

Role of Macrophages and Neutrophils

- Macrophages: 1st line of defense, followed by *granulocytes, neutrophils*
- Move from blood vessels to tissue: *extravasation* 外渗
- Migration of neutrophils:
 - Neutrophil extravasation takes 4 steps:
 - Rolling
 - Activation
 - Arrest
 - Adhesion

Rolling: neutrophils bind briefly to endothelium through weak selectin-carbohydrate interactions

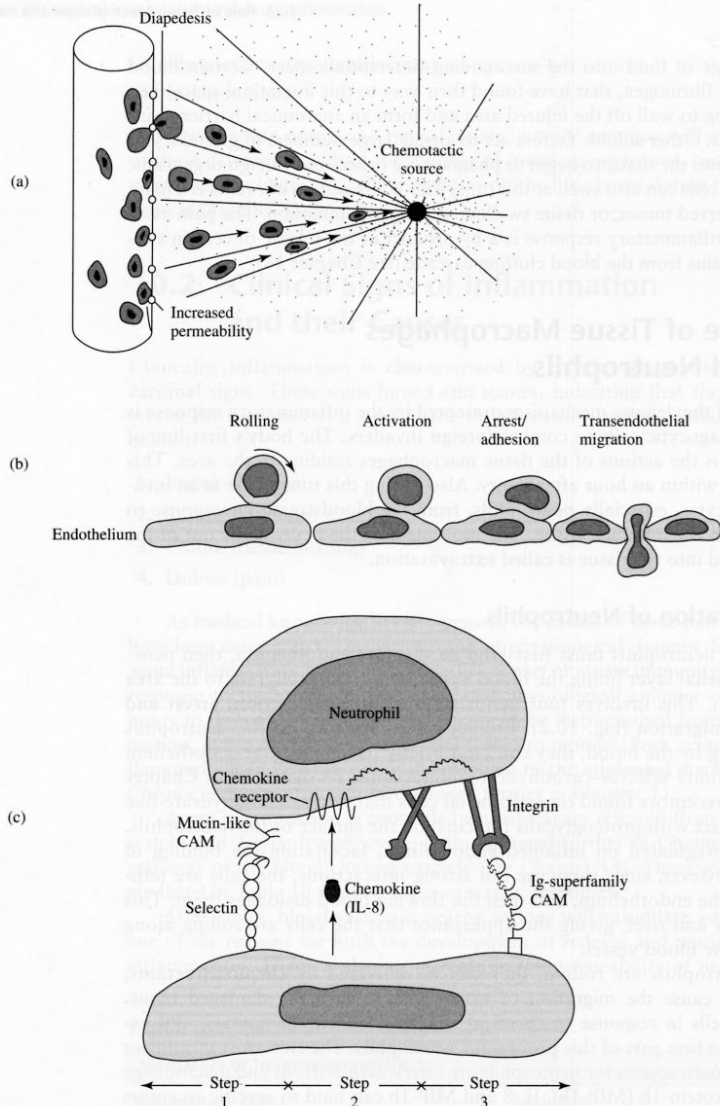
Definition: *Selectins* are receptors on endothelial cells that have a carbohydrate-like portion that binds with proteoglycans (mucins) on neutrophil surface

Activation: neutrophils activated by **chemoattractants**, substances that cause neutrophils to migrate toward site of injury

Definition: *chemotaxis* is the movement of cells in response to chemical stimuli

Arrest/adhesion: *neutrophils* stop rolling and attach to endothelial cells

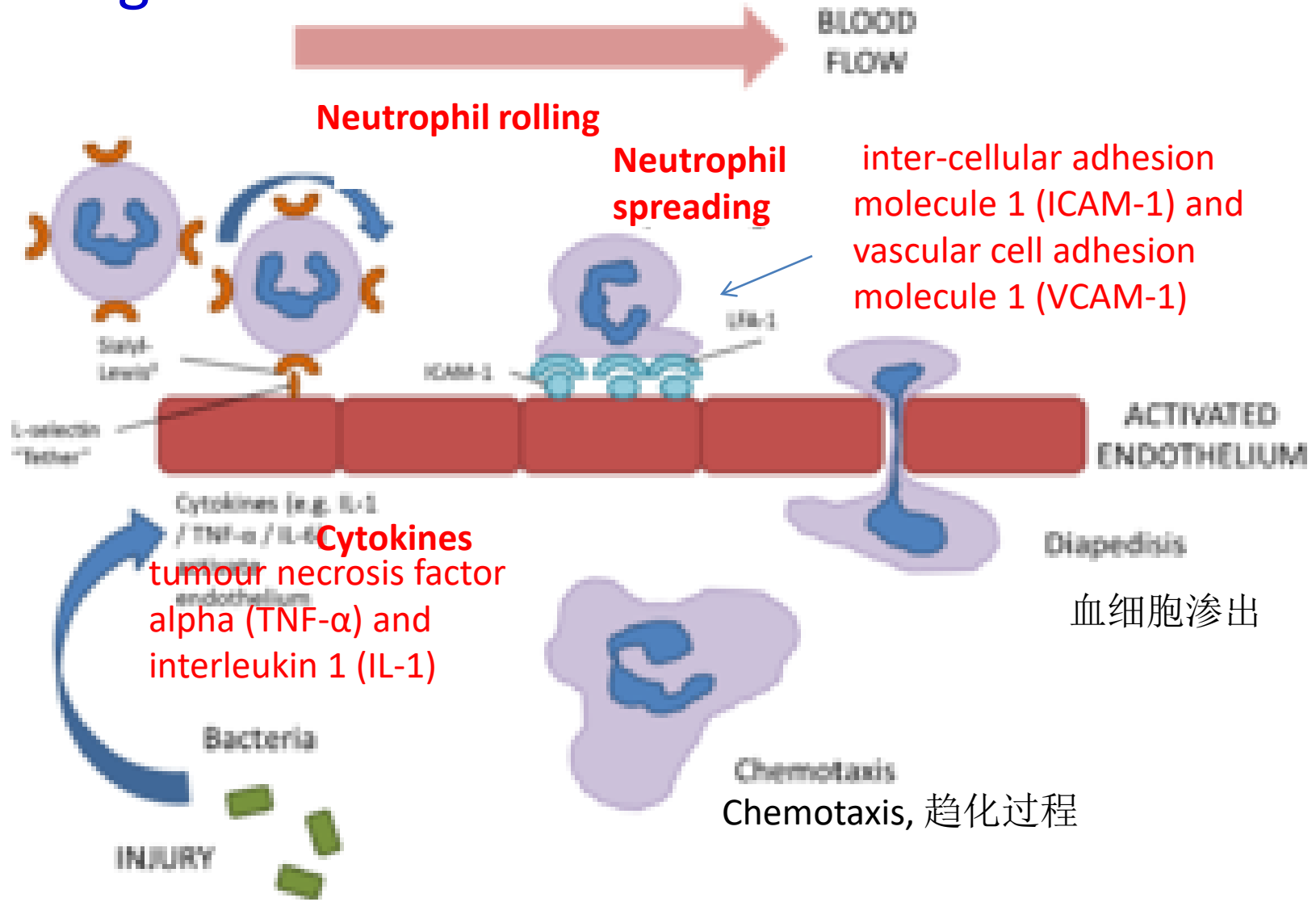
Transendothelial migration: *diapedesis*, or the squeezing of parts of the cell at a time through the endothelial cells



内皮迁移:血细胞渗出

Actions of Neutrophils

Killing via phagocytosis; Respiratory Burst; Release of signaling molecules

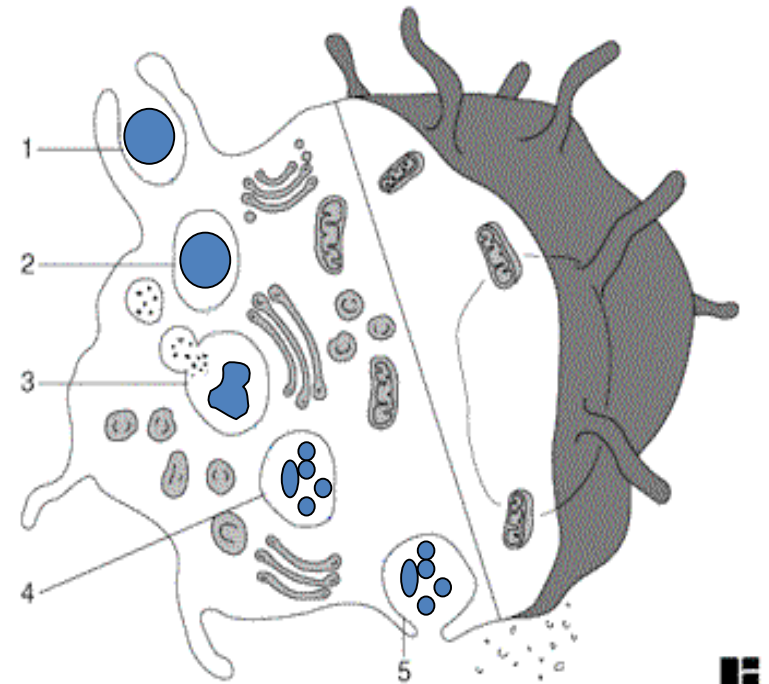


Phagocytosis

- **Phagocytosis:**

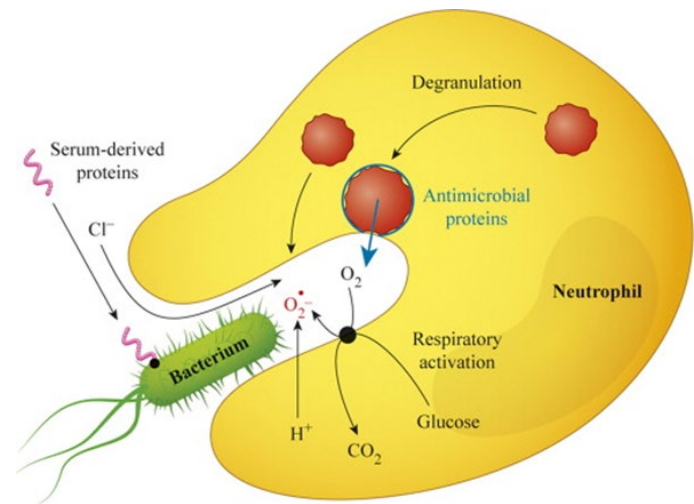
Engulfing and degradation or digestion of fragments of tissue or material

1. long membrane evagination (鼓出), called pseudopodia (伪足).
2. Ingestion forming a "phagosome," which moves toward the lysosome.
3. Fusion of the lysosome and phagosome (phagolysosome), releasing lysosomal enzymes
4. Digestion of the ingested material.
5. Release of digestion products from the cell.



Respiratory Burst

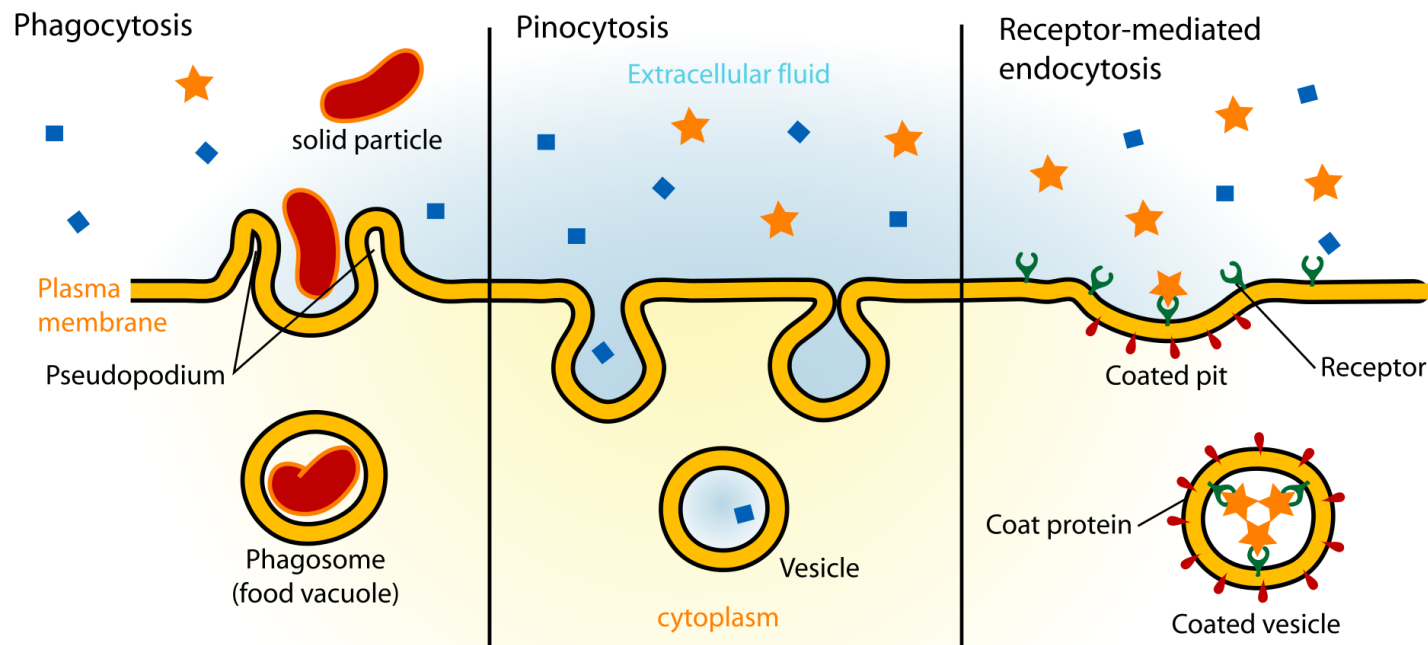
- Glucose metabolism increased 10-fold
- Oxygen consumption increased 2-3 fold
- Formation of reactive oxygen and nitrogen species (*radicals and oxidizers*) to kill foreign invaders
- These substances promote corrosion of biomaterials and may cause unwanted tissue damage



Endocytosis is a cellular process in which substances are brought into the cell. The material to be internalized is surrounded by an area of plasma membrane, which then buds off inside the cell to form a vesicle containing the ingested material.

Endocytosis includes pinocytosis (cell drinking) and phagocytosis (cell eating). It is a form of active transport.

Endocytosis



细胞因子，也翻译为**细胞激素**（[英语](#)：cytokine），是一组[蛋白质](#)及[多肽](#)在生物中用作信号蛋白。这些类似[激素](#)或[神经递质](#)的蛋白用作细胞间沟通的信号。细胞因子多是水溶性蛋白和糖蛋白，分子量小（8-30千[道尔顿](#)）。

细胞因子可以由多种细胞释放，尤其重要的是在[先天性免疫反应](#)和[适应性免疫反应](#)。由于其免疫系统中的作用，细胞因子参与免疫性疾病、炎症及传染性疾病。不过，并非所有的功能仅限于[免疫系统](#)，细胞因子还涉及多个[胚胎发育](#)环节。

细胞因子是由多种细胞类型（如造血性和非造血细胞）产生。并能对邻近细胞或整个机体有作用。这些效应强烈依赖于其他化学因子和细胞因子的存在。

根据其假定功能，来源细胞或靶细胞，细胞因子曾被命名为[淋巴因子](#)、[白细胞介素](#)和[趋化因子](#)。因为细胞因子的特点是相当冗余性和多效性，这些区分已经过时。

白细胞介素一词最初使用是因为有些研究者推定其主要靶细胞是[白细胞](#)。现主要用于命名新的细胞分子，与其可能的功能没有太大的关系。**趋化因子**是指可以介导[细胞趋化](#)的细胞因子。值得注意的是，[白细胞介素-8](#)（Interleukin-8，IL-8）是唯一的趋化因子被命名为白细胞因子。

作用

每个细胞因子与其特异的细胞表面受体结合，由此引起的细胞内信号传递而改变细胞功能。这可能包括上调或下调几个基因的[转录](#)及其[转录因子](#)。从而导致细胞内其它细胞因子的生产、细胞表面受体分子的增加，或通过反馈抑制而抑制其自身的作用。对于某一特定细胞的某一特定细胞因子，其作用取决于细胞外浓度，相应的细胞表面受体的多少，以及配体—受体结合活化后的下游信号。而后两种因素可因细胞类型而异。有趣的是，细胞因子的特点是相当的“过剩”，许多细胞因子似乎有着相似的功能。

细胞因子的功能可以被归纳为：

[自分泌](#)：细胞因子作用于释放它的细胞当中。

[旁分泌](#)：细胞因子作用于相邻的细胞上。

[内分泌](#)：细胞因子扩散到远处的区域（通过血液或血浆）来影响不同组织。

Secretion of Chemical Mediators

- Neutrophils secrete *cytokines*

Definition: *cytokines* are a category of signaling proteins and glycoproteins that, like hormones and neurotransmitters, are used extensively in cellular communication.

TABLE 10.2

Actions of Cytokines	
Interleukin-1 (IL-1)	Promotes granulocyte migration, increases production of IL-8, activates lymphocytes, promotes release of tissue factor
Interleukin-6 (IL-6)	Activates lymphocytes
Interleukin-8 (IL-8)	Attracts neutrophils, promotes neutrophil integrin/CAM interaction
Macrophage inflammatory protein 1a (MIP-1a)	Attracts monocytes
Macrophage inflammatory protein 1b (MIP-1b)	Same effect as IL-8 on neutrophils, attracts monocytes
Transforming growth factor- β (TGF- β)	Limits inflammatory response
Tumor necrosis factor- α (TNF- α)	Promotes granulocyte migration, increases production of IL-8, promotes release of tissue factor

TABLE 10.3

Cytokines Involved in Inflammation/Immunity	
Cytokines produced by inflammatory cells to act on inflammatory cells	IL-1, IL-8, MIP-1a, MIP-1b, TGF- β , TNF- α
Cytokines produced by inflammatory cells to act on lymphocytes	IL-1, IL-6, TNF- α

Role of other *Leukocytes*

Monocytes/macrophages

Monocytes arrive at injury ~6 hrs after inflammatory response begins

They enlarge to form *macrophages*. Macrophages take up to 8 hours to mature and form a large quantity of *lysosomes*, or digestive enzymes.

Macrophage then becomes the dominant type of cell. They are similar to neutrophils, but with greater killing capacity.

Macrophages:

- Phagocytose foreign invaders
- Secrete chemical mediators
- Coordinate response of other body systems
- Act as intermediary between innate and acquired immune response systems

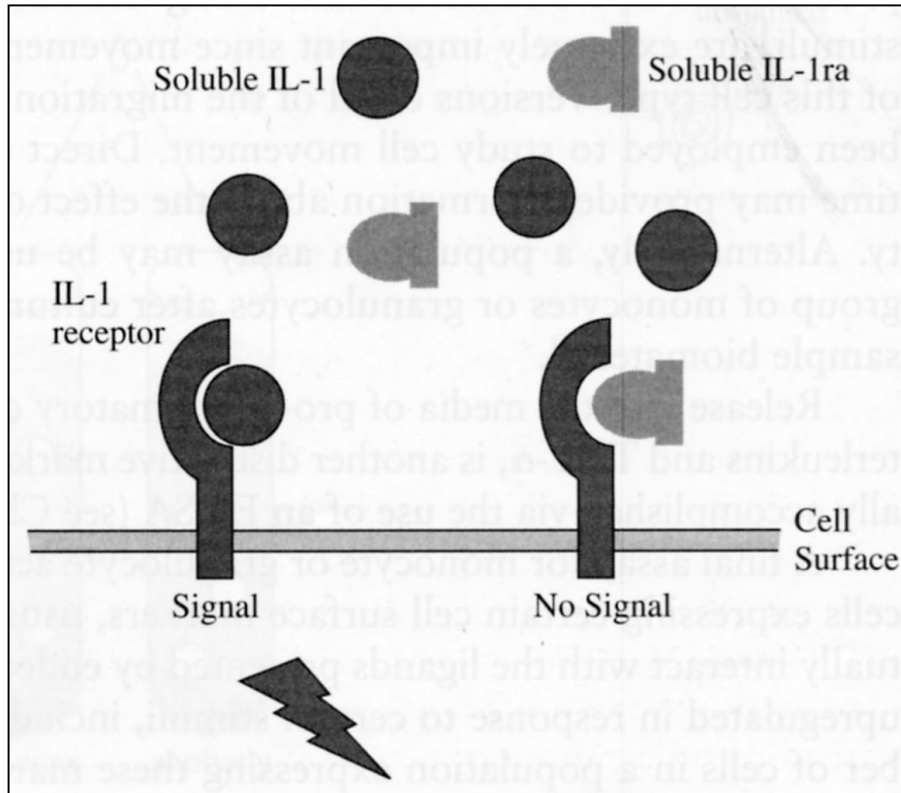
Phagocytosis in Biomaterials

- If phagocytosed material resists degradation it can remain in a macrophage until it dies and undergoes *lysis* (disintegration) and is released
- If macrophages cannot digest particles, *fibroblasts* can form to encapsulate particles
- Objects too big to be ingested result in *frustrated phagocytosis*. Neutrophils and macrophages release *lysosomal* materials (digestive enzymes)

Other Granulocytes

- Eosinophils
 - Respond as neutrophil, but have less phagocytic ability
 - Attach and destroy parasites
 - Prevent spread of inflammation
- Basophils:
 - Release heparin, histamine, bradykinin, serotonin (soluble mediators of inflammation)

Termination of Inflammation



Inflammation ends with release of IL-1ra, a *receptor antagonist* to IL-1. IL-1ra binds to receptors as IL-1 but does not stimulate them.

TGF- α : transforming growth factor α , inhibits certain cell types involved in inflammatory response

IL-1ra and TGF- α must act within a small radius of where they are produced

Summarize the Life Span of Leukocytes